8085 INTRODUCTION

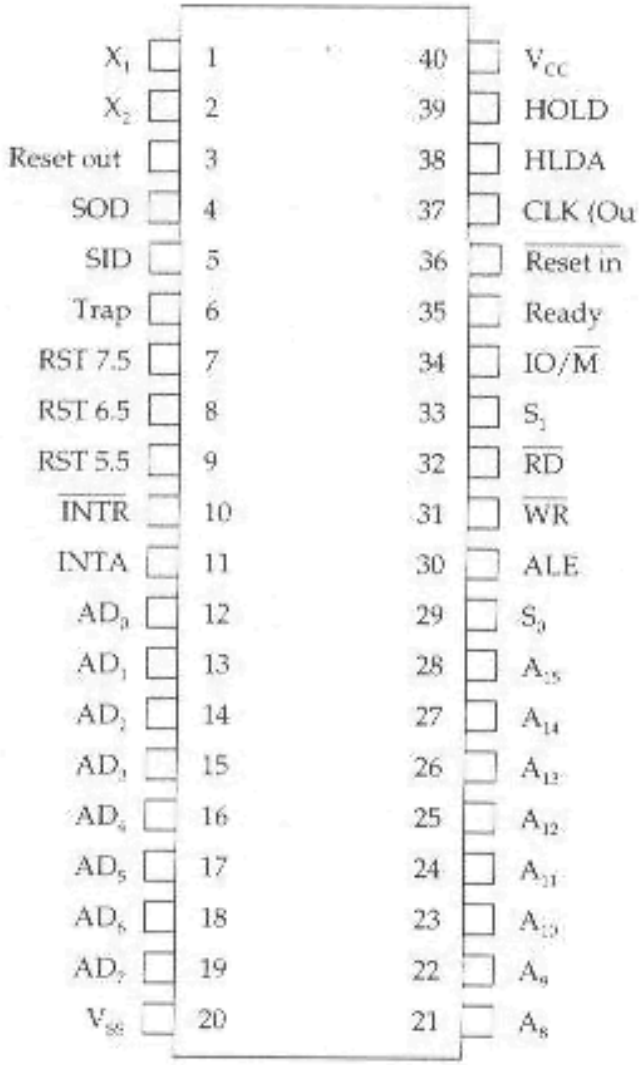
The features of INTEL 8085 are :

- It is an 8 bit processor.
- It is a single chip N-MOS device with 40 pins.
- It has multiplexed address and data bus.(AD_0-AD_7).
- It works on 5 Volt dc power supply.
- The maximum clock frequency is 3 MHz while minimum frequency is 500kHz.
- It provides 74 instructions with 5 different addressing modes.

8085 INTRODUCTION

- It provides 16 address lines so it can access 2^16 =64K bytes of memory.
- It generates 8 bit I/O address so it can access 2^8=256 input ports.
- It provides 5 hardware interrupts:TRAP, RST 5.5, RST 6.5, RST 7.5,INTR.
- It provides Acc ,one flag register ,6 general purpose registers and two special purpose registers(SP,PC).
- It provides serial lines SID ,SOD.So serial peripherals can be interfaced with 8085 directly.

8085 PIN DIAGRAM



8085 PIN DESCRIPTION

Some important pins are :

- AD_0-AD_7: Multiplexed Address and data lines.
- A_8-A_{15}: Tri-stated higher order address lines.
- ALE: Address latch enable is an output signal.It goes high when operation is started by processor .
- S0,S1: These are the status signals used to indicate type of operation.
- RD^{-}: Read is active low input signal used to read data from I/O device or memory.
- WR^{-}:Write is an active low output signal used write data on memory or an I/O device.

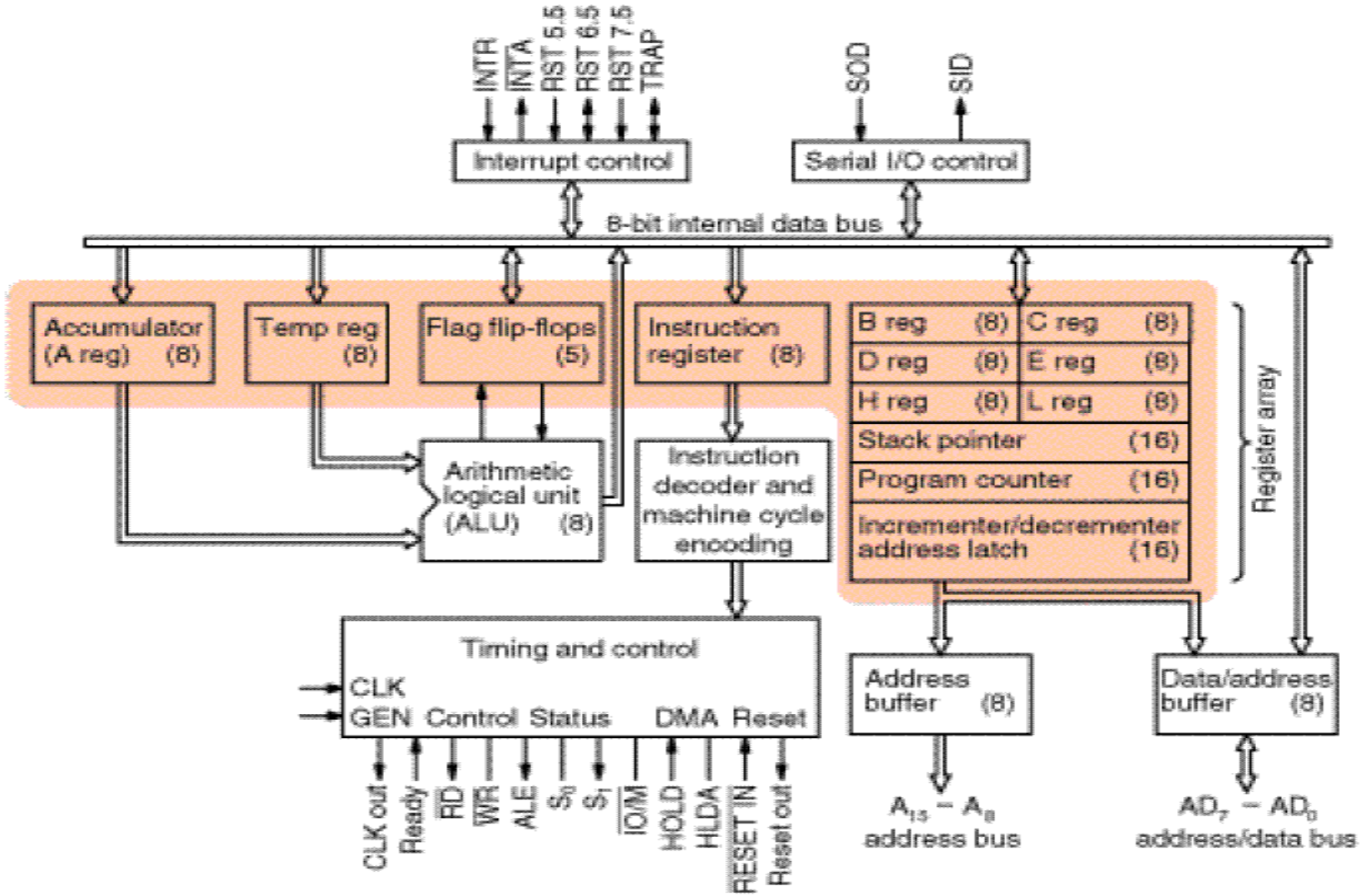
8085 PIN DESCRIPTION

- READY:This an output signal used to check the status of output device.If it is low, µP will WAIT until it is high.
- TRAP:It is an Edge triggered highest priority , non mask able interrupt. After TRAP, restart occurs and execution starts from address 0024H.
- RST5.5,6.5,7.5:These are maskable interrupts and have low priority than TRAP.
- INTR¯&INTA:INTR is a interrupt request signal after which µP generates INTA or interrupt acknowledge signal.
- IO/M¯:This is output pin or signal used to indicate whether 8085 is working in I/O mode(IO/M¯=1) or Memory mode(IO/M¯=0).

8085 PIN DESCRIPTION

- HOLD&HLDA:HOLD is an input signal .When µP receives HOLD signal it completes current machine cycle and stops executing next instruction.In response to HOLD µP generates HLDA that is HOLD Acknowledge signal.
- RESET IN¯:This is input signal.When RESET IN¯ is low µp restarts and starts executing from location 0000H.
- SID: Serial input data is input pin used to accept serial 1 bit data .
- X_1X_2 :These are clock input signals and are connected to external LC,or RC circuit.These are divide by two so if 6 MHz is connected to X_1X_2, the operating frequency becomes 3 MHz.
- VCC&VSS:Power supply VCC=+ -5Volt& VSS=-GND reference.

8085 ARCHITECTURE



Arithmetic and Logical group

Accumulator: It is 8 bit general purpose register.

- It is connected to ALU.
- So most of the operations are done in Acc.

Temporary register: It is not available for user

- All the arithmetic and logical operations are done in the temporary register but user can’t access it.

Flag: It is a group of 5 flip flops used to know status of various operations done.

- The Flag Register along with Accumulator is called PSW

or Program Status Word.

Arithmetic and Logical group

Flag Register is given by:

S	Z	X	AC	X	P	X	CY

S:Sign flag is set when result of an operation is negative.

Z:Zero flag is set when result of an operation is 0.

Ac:Auxiliary carry flag is set when there is a carry out of lower nibble or lower four bits of the operation.

CY:Carry flag is set when there is carry generated by an operation.

P:Parity flag is set when result contains even number of 1's.

Rest are don't care flip flops.

Register Group

- Temporary registers (W,Z):These are not available for user. These are loaded only when there is an operation being performed.
- General purpose:There are six general purpose registers in 8085 namely B,C,D, E,H,L.These are used for various data manipulations.
- Special purpose :There are two special purpose registers in 8085:

1. SP :Stack Pointer.
2. PC:Program Counter.

Register Group

Stack Pointer: This is a temporary storage memory 16 bit register. Since there are only 6 general purpose registers, there is a need to reuse them .

- Whenever stack is to be used previous values are PUSHED on stack and then after the program is over these values are POPED back.

Program Counter: It is 16 bit register used to point the location from which the next instruction is to be fetched.

- When a single byte instruction is executed PC is automatically incremented by 1.
- Upon reset PC contents are set to 0000H and next instruction is fetched onwards.

INSTRUCTION REGISTER,DECODER & CONTROL

- Instruction register:When an instruction is fetched , it is executed in instruction register.This register takes the Opcode value only.
- Instruction decoder: It decodes the instruction from instruction register and then to control block.
- Timing and control:This is the control section of µP.It accepts clock input .

INTERRUPT CONTROL

- It accepts different interrupts like TRAP INT5.5,6.5,7.5and INTR.

SERIAL IO CONTROL GROUP

- It is used to accept the serial 1 bit data by using SID and SOD signals and it can be performed by using SIM & RIM instructions.